Jules DUPONT

Delft, NL | julesdupont02@gmail.com | +33609781351 | LinkedIn | Jules02 | jules02.github.io

Education

Delft University of Technology & Leiden University *Sept 2025 – Present*
MSc Quantum Information Science & Technology

- Double degree, completing my engineering diploma from École des Mines | **Level:** 9.0/10 (first year courses)
- **Relevant coursework:** Fundamentals of Quantum Information, Qubit Dynamics and Quantum Control, Quantum Devices and Interfaces, Quantum Computation and Quantum Programming, Applied Quantum Algorithms, Quantum Computer Architecture, Quantum Error Correction

École Nationale Supérieure des Mines de Nancy *Sept 2023 – Present*
Diplôme d'Ingénieur (MSc level), Major in Computer Science

- 5th best French graduate school of engineering, or *Grande École* (L'Étudiant, 2025)
- Excellence Scholarship – awarded by the alumni foundation (2025)
- **Relevant coursework:** Algorithms and Complexity, Quantum Physics, Foundations of Computing, Computer Architecture, Software Engineering, Introduction to Blockchains, Complex Analysis

Classe Préparatoire aux Grandes Écoles *Sept 2021 – July 2023*
Lycée Saint-Louis (Paris)

- Two-year undergraduate program preparing for entrance exams to French most prestigious *Grandes Écoles*
- **Relevant coursework:** Mathematics, Physics, Computer Science

Projects

Hackathon winner *Hack the World(s)* *June 2026*
24h on world models & JEPA architectures, with the support of Yann Le Cun

- Won 1st place out of 25 teams (100 participants, 600+ applicants)
- Built a V-JEPA model learning the latent dynamics of physical fields (Gray-Scott reaction-diffusion), outperforming a pixel-space baseline on long horizons
- Trained on the Dalia supercomputer (NVIDIA GB200 NVL72)

Research Project *Parcours Recherche at Centre INRIA de Nancy* *Sept 2024 – July 2025*

- Under the supervision of Dr Miriam Backens from the MOCQUA team, worked on Measurement-Based Quantum Computing (MBQC) with the help of the ZX-Calculus
- In parallel with my studies at École des Mines (1.5 days per week)

Research Project *TIPE at Lycée Saint-Louis* *2022 – 2023*

- Studied the use of Support Vector Machines (SVM) in computerized face detection
- Implemented a SVM-based face detection model, with Python

Experience

Joint Interdisciplinary Project *(JIP)* *Remote*
Quobly *Sept 2026 – Nov 2026*

- Implemented Quantum Reservoir Computing (QRC) for credit card fraud detection
- Worked in an interdisciplinary team, alongside students from outside the quantum field

President of the school's IT and cybersecurity club *(15 members)* *Nancy, France*
Cybermines *Sept 2024 – July 2025*

- Ran weekly cybersecurity sessions on TryHackMe
- Organized conferences with industry stakeholders

Software Engineer Internship *Paris, France*
Eres Group *Aug 2024*

- Implemented a module in Java for enabling SMS Code 2FA with Keycloak
- Applied agile software development principles and built with Test driven development (TDD)

Skills

Programming languages: Python, C/C++, Java, JavaScript, x86 assembly, SQL

Machine learning: PyTorch, scikit-learn, NumPy

Quantum computing: Qiskit, PennyLane

Technologies: Linux/Unix, Git, GitHub, Docker, LaTeX

Project management: Agile methods, completed the MOOC *Gestion de Projet* in 2024

Languages

- **French:** Native
- **English:** Proficient (C1, 7.5/9.0 IELTS in June 2024)
- **German:** Independent user (B2 Goethe in February 2025)

Hobbies

- **Sports.** I have been competing in **track running (national level)** since I was 12 years old. At École des Mines, I was the head of the school's athletics team.
- **Music.** At École des Mines, I was a DJ in the school's mix club.

Additional Information

- **Humanitarian Commitment.** During the first year of my studies at École des Mines, I served as vice-president of Burkin'Action, the school's humanitarian club that supports a village in Burkina Faso.
- Since 2023, I hold a French **driving license** (*Permis B*).